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Effects of clonal integration and nitrogen supply on responses of a clonal plant to short-term herbivory

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Abstract

Aims
Clonal integration, i.e. resource sharing between interconnected ramets, can help clonal plants tolerate abiotic stress. However, few studies have directly examined the ecological significance of clonal integration in the ability of clonal plants to tolerate biotic stress such as herbivory.

Methods
We grew clonal fragments of an invasive clonal plant *Alternanthera philoxeroides*, each consisting of an apical part (relatively young) and a basal part (relatively old), at two levels (low and high) of soil nitrogen (N). For each fragment, the apical part was subjected to either no herbivory or herbivory for 2 weeks by a specialist insect, *Agasicles hygrophila*, and was either connected with or disconnected from the basal part not subjected to herbivory.

Important Findings
Herbivory in the apical part severely reduced leaf growth (leaf mass, leaf number or leaf area) and ramet production of both apical and basal parts, and increased root to shoot ratio of the apical part. Irrespective of herbivory and soil N, stem connection between the apical and the basal part decreased root to shoot ratio of the apical part, but did not change its growth measures. Meanwhile, connection increased stem N concentration of the apical part growing under high-N supply and decreased stem carbon (C) concentration of the apical part under low-N supply. By contrast, connection increased root to shoot ratio of the basal part, but reduced its leaf and ramet production. Connection also increased leaf and stem C concentrations of the basal part under low-N supply. Thus, clonal integration can be beneficial for the early development of young *A. philoxeroides* ramets, but not for their local adaptation to herbivory damage by *A. hygrophila*.

Keywords: *Agasicles hygrophila*, *Alternanthera philoxeroides*, foliar herbivory, invasive species, physiological integration

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INTRODUCTION

Clonal integration is a distinct trait of clonal plants, which allows the transport and sharing of internal resources (e.g. photosynthates, water and mineral nutrients) among connected ramets (Alpert and Mooney 1986; Song *et al.* 2013; Yu *et al.* 2004, 2009). In heterogeneous environments, clonal integration often improves the performance of recipient ramets growing in unfavorable patches at no costs to donor ramets growing in favorable patches, thereby enhancing the overall fitness of whole clones (Hutchings and Wijesinghe 1997; Song *et al.* 2013; Wang *et al.* 2017c). A large body of studies have

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demonstrated that clonal integration can improve the survival, growth and competitive ability of clonal plants growing in environments with heterogeneously distributed abiotic resources (e.g., such as light, water and nutrients; Alpert 1996; Hutchings and Wijesinghe 1997; Roiloa *et al.* 2014; Wang *et al.* 2017c; Xu *et al.* 2010). However, only a few studies have explicitly examined the ecological significance of clonal integration under biotic interactions with other trophic levels such as herbivory (Gomez *et al.* 2008, 2010), colonization by arbuscular mycorrhizal fungi (Du *et al.* 2009) or pathogenesis (D'Hertefeldt and van der Putten 1998; Stuefer *et al.* 2004). Such knowledge is important for the understanding of the evolutionary significance of clonal integration.

Clonal integration is expected to enhance the ability of clonal plants to tolerate herbivory for several reasons. First, when parts of a clone are damaged, internal resources in undamaged ramets can be exported *via* horizontal structures such as stolons or rhizomes to damaged ramets, compensating for tissue loss of damaged ramets and alleviating herbivory-induced resource deficiency (Liu *et al.* 2009; Tewari *et al.* 2014). For instance, clonal integration could guarantee sufficient photosynthate supply for the fruit development of the cultivated cranberry *Vaccinium macrocarpon* injured by the larvae of *Dasineura oxycoccana* (Tewari *et al.* 2014). Second, clonal integration can induce functional specialization among connected ramets (Lyu *et al.* 2016; You *et al.* 2014). One predictable scenario is that ramets damaged by aboveground herbivory specialize in the acquisition of belowground resources (e.g. mineral nutrients or water; Lyu *et al.* 2016; You *et al.* 2014), and undamaged ramets specialize in the utilization of aboveground resources (e.g. sunlight). Such functional specialization is to some extent similar to division of labor in clonal plants growing in heterogeneous light environments (e.g. Alpert *et al.* 2003; Lin *et al.* 2018; Roiloa *et al.* 2014; Xu *et al.* 2010). Third, after herbivore damage, clonal integration may contribute to the regrowth of damaged ramets (Gao *et al.* 2014; Hosaka *et al.* 2016; Liu *et al.* 2009; Wang *et al.* 2017b). Several studies have reported that clonal integration could facilitate compensatory growth of damaged ramets of clonal plants (Gao *et al.* 2014; Lyu *et al.* 2016; Schmid *et al.* 1988; Wang *et al.* 2017b; You *et al.* 2014). However, we still need empirical work testing the mechanism by which clonal plants tolerate herbivory.

In addition, the effects of clonal integration on tolerance of clonal plants to herbivory may be modified by external resource availability (Gao *et al.* 2014). The compensatory continuum hypothesis predicts that plant tolerance to herbivory may be constrained by resource limitation, i.e. plants inhabiting resource-poor environments are often less tolerant to herbivory than those growing in resource-rich environments (Fornoni *et al.* 2003; Hawkes and Sullivan 2001; Wise and Abrahamson 2007). One plausible extrapolation is that tolerance to herbivory mediated by clonal integration may also be weakened by resource limitation (Gao *et al.* 2014). For clonal plants growing in habitats with limited resources, clonal integration may improve the fitness of damaged (recipient) ramets, but may also equally inhibit the performance of undamaged (donor) ramets, so that there may be zero net effect on the whole clone (Dong *et al.* 2015). By contrast, for clonal plants in habitats where resources are unlimited, the costs to undamaged ramets caused by clonal integration may be mitigated due to increased resource availability, so that clonal integration brings more benefits to damaged ramets and thus to the whole clone (Dong *et al.* 2015; Gao *et al.* 2014). Although effects of resource availability on plant tolerance to herbivory have been extensively studied (Coley *et al.* 1985; Gianoli and Salgado-Luarte 2017; Sun *et al.* 2009, 2010; Wise and Abrahamson 2007), little is known about the interactive effects between clonal integration and resource availability on herbivory tolerance of plants.

Accordingly, in this study, we tested the effects of clonal integration and nitrogen (N) supply on the growth and physiological responses of the invasive clonal species *Alternanthera philoxeroides* to foliar herbivory by the specialist beetle *Agasicles hygrophila*. Specifically, we tested the following hypotheses: (i) Clonal integration can improve the performance of ramets damaged by herbivory. (ii) Clonal integration can cause a fitness cost to connected undamaged ramets. These two hypotheses are based on the premise that clonal integration allows resource sharing between connected ramets that experienced contrasting herbivory conditions, thereby resulting in a compensatory effect for the tissue loss of damaged ramets, but also causing a decline in resource availability for connected undamaged ramets. (iii) Clonal integration can increase the overall fitness of clones under herbivory, if the benefits of integration to damaged ramets overwhelm the costs to undamaged ramets. (iv) The consequence of clonal integration related to herbivory becomes more beneficial with increasing resource availability. This is because the costs of clonal integration may be reduced as external resource availability increases.

MATERIALS AND METHODS

Plant and insect species

Alternanthera philoxeroides (Mart.) Griseb (Amaranthaceae) is a creeping, perennial herb that is native to South America (Holm *et al.* 1997; Sainty *et al.* 1998) and is listed as one of the worst invasive weeds in many countries, including China, the USA, New Zealand and Australia (Holm *et al.* 1997; Julien *et al.* 1995). In China, *A. philoxeroides* rarely reproduce sexually and, instead, produces offspring ramets mainly through the fragmentation of stems and roots (Xu *et al.* 2003; Ye *et al.* 2003). The species occupies a wide range of terrestrial and aquatic habitats, such as crop fields and irrigation ditches (Pan *et al.* 2006; Sainty *et al.* 1998). Clonal integration has been reported to facilitate the invasion of *A. philoxeroides* (Wang *et al.* 2008, 2009; Xu *et al.* 2010; Yu *et al.* 2009).

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The flea beetle *Agasicles hygrophila* Selman and Vogt (Coleoptera: Chrysomelidae) is a specialist herbivore of *A. philoxeroides* (Spencer and Coulson 1976), which was introduced to China since 1986 as a biological control agent against *A. philoxeroides* (Lu and Ding 2012). Larvae and adults of *A. hygrophila* feed on leaves and stem buds (Spencer and Coulson 1976), and third-stage larvae pupate in hollow stem internodes of *A. philoxeroides* (Fu *et al.* 2016; Pan *et al.* 2011).

Plant materials were collected from three populations in a riparian agricultural area on 18–19 May 2011 in Taizhou, Zhejiang Province, China (28.87°N, 121.01°E). To circumvent the confounding effects of maternal environments, the materials were mixed and then cultivated through vegetative propagation for 6 years in a greenhouse belonging to the Forest Science Company, Ltd., of Beijing Forestry University. Individuals of *A. hygrophila* were collected from the same agricultural area in May 2017.

Experimental design

On 23 April 2017, 70 clonal fragments, each comprising a single stem (~25 cm long) with four stem nodes and an apex, were transplanted into pairs of adjacent plastic pots. For each fragment, the two younger nodes with the stem apex were placed in one pot, the two older nodes were placed in the adjacent pot, and the stem internode connecting the young and old nodes was run through a hole (1.3 cm in diameter) in the adjoined sides of the two pots. The pots were 14 cm in diameter and 12 cm deep and were filled with a 1:1 (v:v) mixture of quartz sand (0.5–1 mm particle size) and peat (Pindstrup Seedling; Pindstrup Mosebrug A/S, Pindstrup, Denmark).

After 5 weeks when all the four initial stem nodes of each fragment had rooted and produced new stems, 48 clonal fragments were selected for experiment use and 6 additional fragments were used for initial measurements. In each clonal fragment, the two younger initial nodes and their newly produced stems were termed as the apical part, and the two older initial nodes and their newly produced stems were termed as the basal part. The initial mass of the apical and the basal part were 1.67 ± 0.17 and 0.30 ± 0.07 g (mean ± 1 *SE*, $n = 6$), respectively, and the initial number of ramets (nodes) of the apical and basal parts were 15.8 ± 2.0 and 5.2 ± 1.6 (mean ± 1 *SE*, $n = 6$), respectively.

The experiment employed a three-factorial design, including foliar herbivory (with or without herbivory by *A. hygrophila*), clonal integration (intact or severed) and N supply (high or low) (Fig. 1). The apical parts of half of the clonal fragments ($n = 24$) were grown either with or without herbivory by *A. hygrophila* (two adult females and two adult males per plant), whereas the basal parts of all 48 clonal fragments were grown without herbivory (Fig. 1). To prevent the beetles from escaping, the apical and basal parts of each clonal fragment were placed in separate insect cages (25 cm long × 25 cm wide × 50 cm high), which were covered with 0.25-mm nylon mesh. For the treatments with clonal integration, the stem internode that connected the apical and the basal part of each clonal

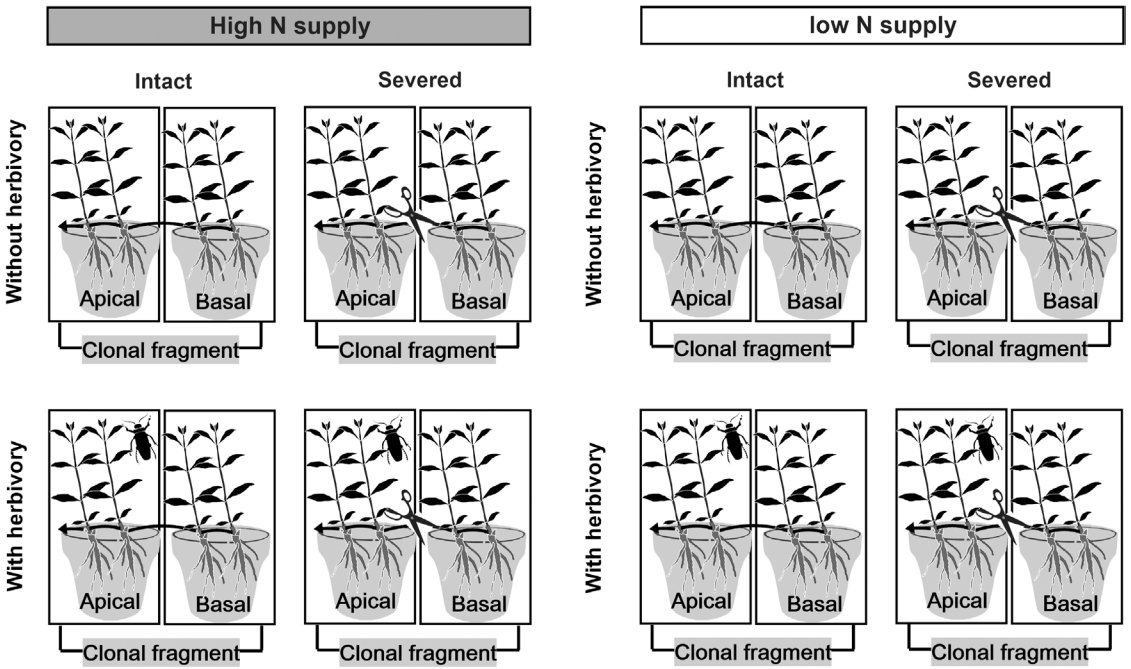

Figure 1: schematic representation of the experimental design. The clonal fragments of *Alternanthera philoxeroides*, each consisting of two basal ramets and two apical ramets with a stem apex (the horizontal arrow), were planted under high-N supply (left panel) and low-N supply (right panel), respectively. The apical parts of clonal fragments were grown either without (top) or with (bottom) the herbivory by *Agasicles hygrophila* and kept either connected (intact, left in each panel) or disconnected (severed, right in each panel) to basal ramets not subjected to herbivory.

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fragment was left intact (n = 24) or severed (n = 24; Fig. 1). Furthermore, to control nutrient supply, the clonal fragments were supplied with modified Hoagland solution with either high (n = 24) or low (n = 24) levels of N (40 or 120 mg N/l, respectively), which was supplied using $Ca(NO_3)_2$ to replace some of the original $CaSO_4$ (online supplementary Table S1; Alpert *et al.* 2002; Wang *et al.* 2017a). The concentrations of N were determined to represent the limiting and non-limiting conditions for *A. philoxeroides*, based on our previous work (Wang *et al.* 2017a). Each pot was irrigated with 500 ml of Hoagland solution for every 3 days. In total, each treatment combination was represented by six replicates, and each replicate was also treated as a block.

The experiment lasted for 2 weeks, from 28 May 2017 to 11 June 2017. During the experiment, the mean temperature and relative humidity were maintained at 24.53 ± 0.64°C and 67.44 ± 2.59%, respectively, which were monitored by using Hobo Temp/RH loggers (HOBO UX100-003; Onset Computer Co., Bourne, MA).

Growth and physiological measurements

After separately harvesting the apical and basal parts of each clonal fragment, the nodes and leaves were counted, and leaf area was estimated using ImageJ v. 1.51j8 (Rasband 2017). The leaves, stems and roots of apical and basal parts were separated, dried at 70°C for 48 h and weighed. Leaf mass, stem mass and root mass were separately determined for the apical part, the basal part and the whole fragment; total mass was calculated as the sum of leaf, stem and root mass; root to shoot ratio was calculated by dividing root mass to shoot mass (i.e. leaf plus stem mass). To measure C and N concentrations, the dried leaves, stems and roots of each plant were finely ground using a Retsch MM400 Mixer Mill (Retsch GmbH, Haan, Germany), with a frequency of 28 Hz for 6 min. The C concentration of each organ was then determined by complete combustion, using a total organic carbon analyzer (Multi N/C 3100, Analytik Jena, Jena, Germany), and the N concentration of each organ was determined using a continuous-flow injection auto-analyzer (Technicon AA3-HR; SEAL Analytical, Norderstedt, Germany).

Data analysis

After ensuring that all data met the assumptions of homoscedasticity and normality, three-way blocked analyses of variance (ANOVAs) were used to test the effects of foliar herbivory (with or without herbivory), clonal integration (intact or severed), N supply (high or low) and their interactions on the growth (total mass, leaf mass, stem mass, root mass, node number, leaf number and leaf area) and biomass allocation (root to shoot ratio) of the apical part, the basal part and the whole fragment (i.e. apical plus basal part), using the 'Anova' function in the CAR package in R v. 3.3.2 (R Core Team 2016). Three-way blocked ANOVAs were also conducted to test the effects of foliar herbivory, clonal integration, N supply and their interactions on physiological measures (leaf C, stem C, root C, leaf N, stem N and root N) of both apical and basal parts. When significant interactions were detected, *post hoc* interaction analysis was followed using the 'testInteractions' function of the *phia* package v. 0.2–1 (Rosario-Martinez 2015).

RESULTS

Performance of apical parts

Irrespective of the herbivory and N-supply treatments, the maintenance of stem connection did not influence growth measures of apical parts, but decreased their root to shoot ratio (Table 1; Fig. 2). Compared with the control, herbivory decreased leaf mass, node number and leaf number, and increased root to shoot ratio of apical parts (Table 1; Fig. 2b, e, f and h). Besides, the negative effects of herbivory on leaf area were also found in the apical parts growing under high-N supply, but not in those under low-N supply (Table 1; Fig. 2g). Under high-N supply, herbivory reduced leaf area of apical parts by 43%, whereas under low-N supply herbivory only reduced leaf area by 28% (Fig. 2g). Compared with the low-N supply, the high-N supply increased total mass, leaf mass, node number, leaf number and leaf area, but decreased root to shoot ratio of apical parts (Table 1; Fig. 2a, b, e, f, g and h). There were no main or interactive effects of herbivory, clonal integration or N supply on stem or root mass of apical parts (Table 1; Fig. 2c and d).

The maintenance of stem connection decreased leaf C concentration of apical parts, irrespective of the herbivory and N-supply treatments (Table 3; Fig. 3a). However, only for the apical parts growing under high-N supply, the maintenance of stem connection increased N concentration of stems (Table 3; Fig. 3e). For the apical parts growing under low-N supply, the maintenance of stem connection increased C concentration of stems, but decreased N concentration of roots (Table 3; Fig. 3b and f). Compared with the control, herbivory increased N concentration of roots, but did not influence other physiological measures (Table 3; Fig. 3). Compared with the low-N supply, the high-N supply increased N concentrations of leaves, stems and roots of apical parts, but decreased C concentration of stems (Table 3; Fig. 3b, d, e and f).

Performance of basal parts

The maintenance of stem connection decreased leaf mass, node number, leaf number and leaf area, but increased root to shoot ratio of basal parts (Table 1; Fig. 4b, e, f, g and h). Compared with the control, herbivory decreased node number and leaf area of basal parts (Table 1; Fig. 4e and g). Compared with the low-N supply, the high-N supply increased total mass, leaf mass, node number, leaf number and leaf area, but decreased root to shoot ratio (Table 1; Fig. 4a, b, e, f, g and h).

Only for the basal parts growing under low-N supply, the maintenance of stolon connection increased C concentration of leaves and stems (Table 3; Fig. 5a and b). Compared with the low-N supply, the high-N supply decreased C concentration of

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Table 1: effects of foliar herbivory (by *Agasicles hygrophila*), clonal integration, N supply and their interactions on the growth and biomass allocation of the apical and the basal parts of clonal fragments of *Alternanthera philoxeroides*

	Total mass	Leaf mass	Stem mass	Root mass	Number of nodes	Number of leaves	Leaf area	Root to shoot ratio
Apical part								
Block	1.8	1.8	2.2	1.1	**4.3****	1.7	1.9	**2.7***
Herbivory (H)	3.1	**9.5****	3.5	0.1	**22.4*****	**26.3*****	**26.8*****	**4.6***
Connection (C)	0.5	0.2	<0.1	2.1	0.6	0.7	<0.1	**5.3***
Nitrogen (N)	**4.3***	**50.8*****	2.2	0.4	**63.3*****	**49.5*****	**48.1*****	**31.3*****
H × C	0.2	0.8	0.1	<0.1	0.6	0.3	<0.1	0.1
H × N	0.1	1.4	<0.1	<0.1	0.3	2.6	**7.3***	<0.1
C × N	<0.1	0.4	<0.1	0.1	0.6	2.0	0.4	0.3
H × C × N	1.2	2.8	1.4	0.1	0.5	0.4	1.3	2.2
Basal part								
Block	**3.1***	**3.2***	**4.4****	1.7	**4.0****	1.7	**3.5***	**2.5***
Herbivory (H)	1.9	2.2	1.6	1.3	**5.6***	2.4	**4.4***	0.4
Connection (C)	1.7	**6.0***	1.8	0.1	**7.5****	**4.4***	**7.4***	**10.2***
Nitrogen (N)	**9.1****	**42.8*****	4.1	0.2	**33.3*****	**34.3*****	**54.9*****	**37.4*****
H × C	0.1	<0.1	0.2	<0.1	<0.1	0.2	<0.1	<0.1
H × N	<0.1	1.4	<0.1	0.4	3.4	0.8	3.8	<0.1
C × N	0.8	<0.1	1.8	1.2	0.2	0.1	0.1	3.3
H × C × N	<0.1	<0.1	<0.1	0.1	0.1	0.7	<0.1	<0.1

Numbers are ANOVA *F*-values. The degrees of freedom (*df*) for block effect is 5,35; the *df* values for foliar herbivory, clonal integration, nitrogen supply and their interactions are all 1,35. Values for which $P < 0.05$ are in bold.
Asterisks indicate significance level: no symbol, $P > 0.05$; *$P = 0.01$–0.05; **$P = 0.001$–0.01; and ***$P < 0.001$.

leaves, but increased N concentrations of leaves, stems and roots (Table 3; Fig. 5a, d, e and f).

Performance of whole clonal fragments

The growth pattern of whole fragments was consistent with that of apical parts, except that the maintenance of stem connection did not influence root to shoot ratio of whole fragments (Tables 1 and 2; Fig. 6). In addition, under high-N supply, herbivory reduced on average 40% of leaf area of whole fragments, whereas under low-N supply, herbivory only reduced by 24% (a significant interactive effect of H × N on leaf area in Table 2; Fig. 6g).

DISCUSSION

Local responses of apical ramets to herbivory

Even in a short experimental period (2 weeks), foliar herbivory by *A. hygrophila* had already exerted detrimental impacts on leaf growth (i.e. leaf mass, leaf number and leaf area) and ramet production of apical parts. These results suggest that *A. hygrophila* could severely inhibit and efficiently control aboveground growth of the host plant, due to their preferential consumption for young leaves of *A. philoxeroides* and their relatively high consumption rates (Sainty *et al.* 1998; Schooler *et al.* 2006).

Our results also show that the maintenance of connection influenced the physiological property of apical ramets, as indicated by the changes in C and/or N concentrations of stems and roots. However, inconsistent with the first hypothesis, such effects of clonal integration did not further contribute to the ability of apical ramets to tolerate foliar herbivory. We speculate that herbivory-induced clonal integration was not triggered because it appeared to be a chronic physiological response of *A. philoxeroides* and only played a key role in the post-herbivory compensatory growth. By contrast, the benefits of clonal integration have been reported in the damaged ramets of clonal plants, during 8–24 weeks after herbivore damage (Gao *et al.* 2014; Liu *et al.* 2009; Lyu *et al.* 2016; Wang *et al.* 2017b; You *et al.* 2014). Another plausible explanation is that the intensity of herbivory was too high in the experiment to obscure any positive effects of clonal integration. At the harvest time of our experiment, the individual density of *A. hygrophila* increased dramatically due to mating activity of adults, and on average, 0.8 adults and 22.6 offspring-generation larvae ultimately survived (first- and second-instar larvae; online supplementary Table S2). Therefore, it is likely that the combined feeding pressure from larvae and adults of *A. hygrophila* resulted in an over-consumption of aboveground organs, again obscuring any benefits of clonal integration.

On the other hand, as an alternative response to foliar herbivory, the apical ramets could maintain a persistent root growth, increase root to shoot ratio and elevate the N content in roots. Given that the root part of *A. philoxeroides* often functions as a belowground storage organ that governs

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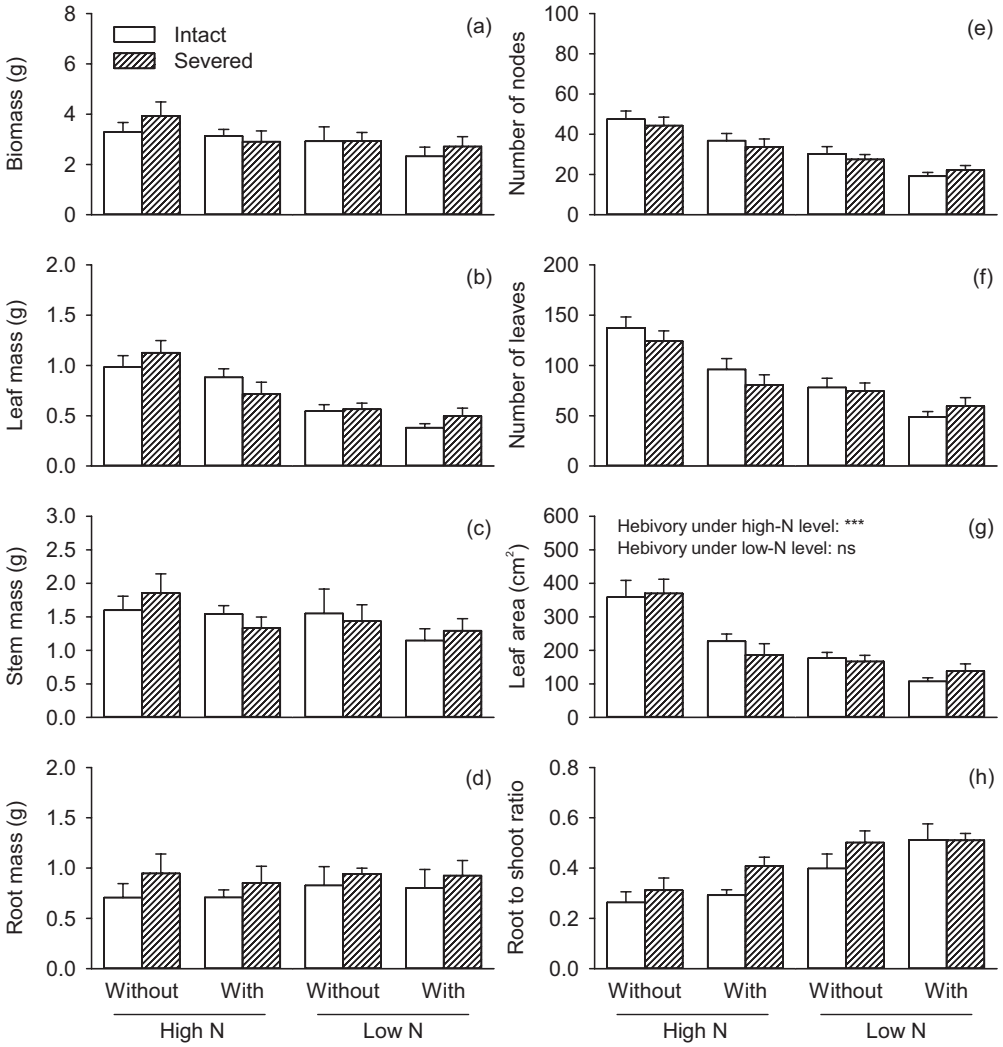

Figure 2: effects of foliar herbivory (with or without herbivory), clonal integration (intact or severed) and N supply (high or low) on the growth measures and biomass allocation of the apical parts of clonal fragments of *Alternanthera philoxeroides*. Error bars indicate means + SE. Symbols indicate significant differences between herbivory treatments at each N level, i.e. $P > 0.05^{ns}$; $P < 0.001$***.

the dormancy of vegetative buds and the formation of new ramets, the maintenance of root development can potentially accelerate their compensatory growth after the loss of above-ground organs caused by foliar herbivory or other disturbances (e.g. trampling, grazing or mechanical shoot removal) and thus greatly benefit the post-herbivory performance of *A. philoxeroides* (Dong *et al.* 2017; Jia *et al.* 2009; Lu *et al.* 2014; Wilson *et al.* 2007).

Non-local responses of basal ramets to herbivory

Although the basal ramets were not directly exposed to herbivory by *A. hygrophila*, our results indicated that herbivory in apical ramets also caused negative effects on leaf growth and ramet reproduction of basal ramets. The results are basically consistent with the second hypothesis and imply that the maintenance of clonal integration causes the fitness costs to donor ramets. The costs to basal ramets might be caused by the export of internal resources to apical ramets (Xu *et al.* 2010) and increased respiration consumption via maintaining inter-ramet connections (Pauliukonis and Gough 2004), and potential transference of phytotoxic elements (Outridge and Hutchinson 1990). Similar results have also been reported in a recently published work of *Carpobrotus edulis*, where the infestation by white garden snails (*Theba pisana*) in basal ramets could significantly cause a reduced growth of undamaged apical ramets (Rodríguez *et al.* 2018).

Responses of whole fragments to herbivory

The negative effects of herbivory in apical ramets ultimately resulted in a decline in the overall fitness of whole fragments. Meanwhile, inconsistent with the third hypothesis, the maintenance of stem connection was unable to improve the fitness of the whole fragments subjected to herbivory because no benefits of clonal integration occurred in the apical ramets

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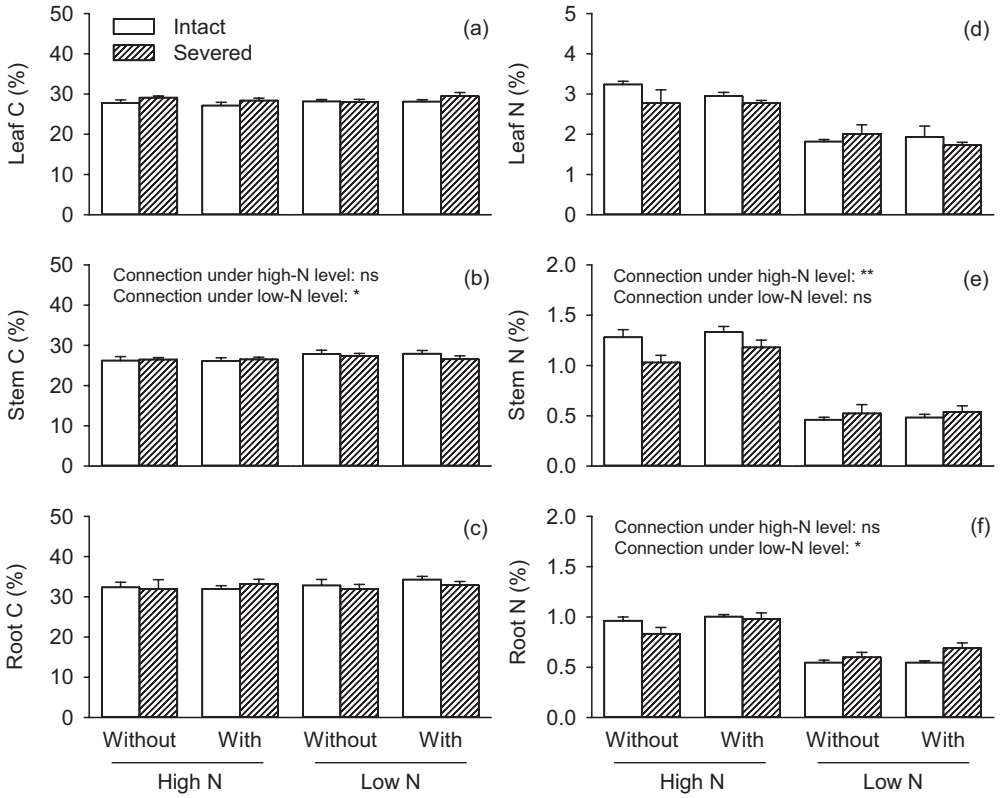

Figure 3: effects of foliar herbivory (with or without herbivory), clonal integration (intact or severed) and N supply (high or low) on the C and N concentrations of different plant organs from the apical parts of clonal fragments of *Alternanthera philoxeroides*. Error bars indicate means + SE. Symbols indicate significant differences between connection treatments at each N level, i.e. $P > 0.05^{ns}$; $P < 0.05$*; $P < 0.01$**.

(Xu *et al.* 2012). The results imply that clonal integration may not be a rapid adaptive response of *A. philoxeroides* to herbivory at the whole-fragment scale. Similarly, clonal integration also contributed less to the overall performance of whole clones of *C. edulis* subjected to herbivory, even though the reciprocal exchange of resources from different specialized ramets occurred (Rodríguez *et al.* 2018).

Roles of clonal integration irrelevant to herbivory

Our results do not support the fourth hypothesis that the consequence of clonal integration related to herbivory becomes more beneficial as the resource availability increases. In fact, we found that the effects of clonal integration on physiological propriety of individual ramets were not related to herbivory, but modified by soil N availability. First, the maintenance of inter-ramet connection improved N contents of stems of apical ramets growing under the high-N condition, although it did not affect that of basal ramets under the same condition. The results suggest that the transportation of N between apical and basal ramets tended to be acropetal toward younger ramets and possibly driven by an internal source-sink relationship (i.e. variation in the N uptake abilities between apical and basal ramets) (Alpert *et al.* 2002; Dong *et al.* 2015; Roiloa *et al.* 2013; Stuefer 1998). Such developmental-programmed clonal integration has been reported to be important for survival and early performance of young ramets in many clonal plants such as *Fragaria chiloensis* and *C. edulis* (Alpert 1996; Alpert *et al.* 2002; Roiloa *et al.* 2013). Besides, the stems of young ramets may also had a higher priority to utilize N-based resources than did other plant organs, which facilitated the rapid expansion of *A. philoxeroides* under favorable environments. Second, the maintenance of clonal integration elevated C contents of both apical and basal ramets growing under unfavorable environments. The results imply that the sharing of C-based resources between apical and basal ramets might be bidirectional, to alleviate their demand for photosynthates under unfavorable habitats (Alpert and Mooney 1986; Alpert *et al.* 2002). Therefore, *A. philoxeroides* may have a flexible ability to regulate the physiological property of both apical and basal ramets under different resource conditions, e.g. developmental-programmed clonal integration.

CONCLUSION

Our work provides preliminary evidence that the maintenance of belowground growth, rather than resource sharing between connected ramets, might be a primary strategy for *A. philoxeroides* to tolerate the transient (2-week) foliar herbivory by *A. hygrophila*. The persistent root growth could potentially improve the local tolerance and post-herbivory

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Table 2: effects of foliar herbivory (by *Agasicles hygrophila*), clonal integration, N supply and their interactions on the growth and biomass allocation of whole clonal fragments of *Alternanthera philoxeroides*

	Total mass	Leaf mass	Stem mass	Root mass	Number of nodes	Number of leaves	Leaf area	Root to shoot ratio
Block	1.9	1.6	**2.7***	1.0	**3.2***	1.6	2.5	**3.0***
Herbivory (H)	3.5	**7.3***	3.9	0.3	**17.3*****	**12.4****	**24.8*****	**4.8***
Connection (C)	1.0	1.9	0.2	1.3	2.0	1.2	1.1	0.5
Nitrogen (N)	**6.8***	**58.8*****	3.5	0.5	**66.8*****	**55.7*****	**74.3*****	**66.3*****
H × C	0.2	0.4	0.2	0.1	0.2	0.3	<0.1	0.2
H × N	0.1	1.7	<0.1	<0.1	2.4	1.9	**8.8****	0.1
C × N	0.1	0.1	0.1	<0.1	0.6	0.9	0.1	1.3
H × C × N	0.7	1.1	1.3	<0.1	<0.1	0.1	0.7	3.0

Numbers are ANOVA *F*-values. The degrees of freedom (*df*) for block effect is 5,35; the *df* values for foliar herbivory, clonal integration, nitrogen supply and their interactions are all 1,35. Values for which $P < 0.05$ are in bold.
Asterisks indicate significance level: no symbol, $P > 0.05$; *$P = 0.01–0.05$; **$P = 0.001–0.01$; and ***$P < 0.001$.

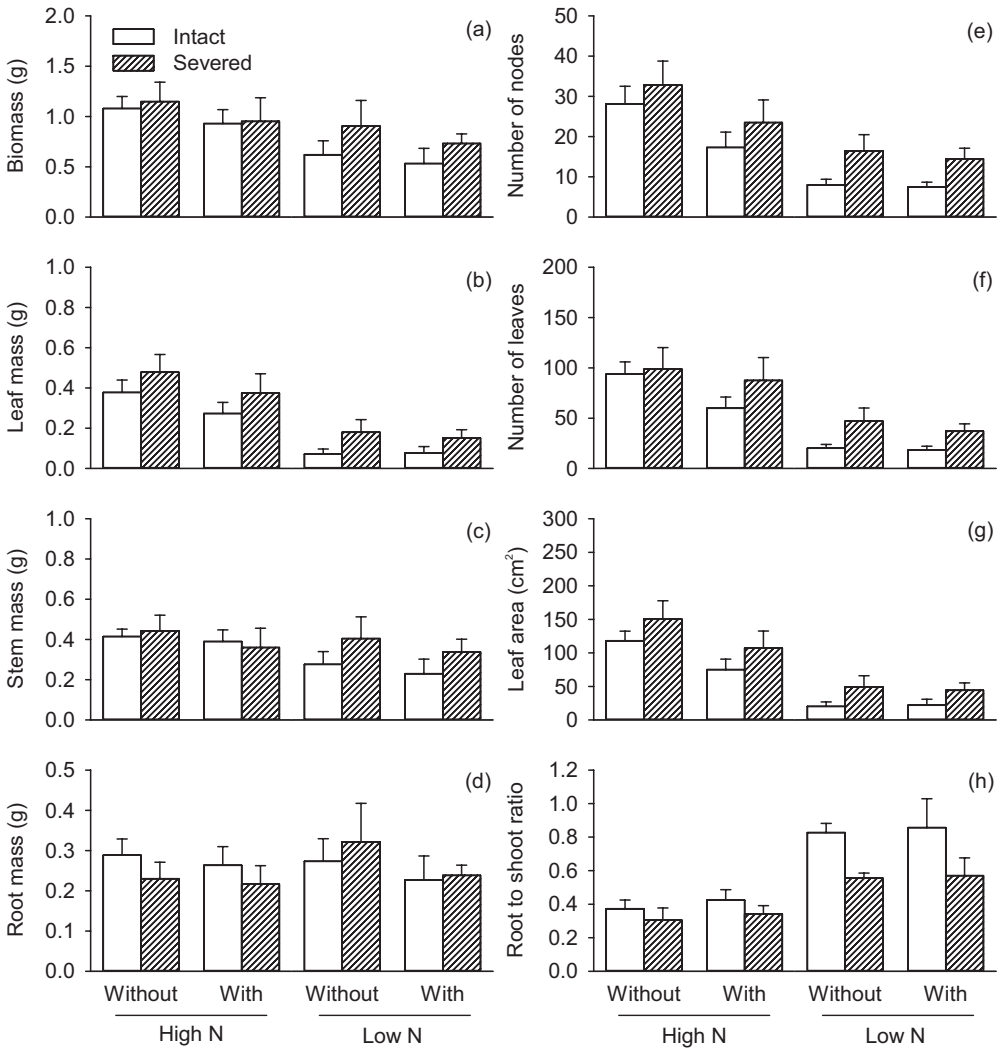

Figure 4: effects of foliar herbivory (with or without herbivory), clonal integration (intact or severed) and N supply (high or low) on the growth measures and biomass allocation of the basal parts of clonal fragments of *Alternanthera philoxeroides*. Error bars indicate means + SE.

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Table 3: effects of foliar herbivory (by *Agasicles hygrophila*), clonal integration, N supply and their interactions on the C and N concentrations of plant organs in the apical and the basal parts of clonal fragments of *Alternanthera philoxeroides*.

	Leaf C	Stem C	Root C	Leaf N	Stem N	Root N
Apical part						
Block	**3.0***	**20.8*****	**8.2*****	1.6	0.4	0.2
Herbivory (H)	<0.1	0.5	1.6	0.9	1.7	**4.6***
Connection (C)	**5.4***	1.3	0.3	1.8	2.3	0.1
Nitrogen (N)	0.7	**14.5*****	0.8	**74.1*****	**228.0*****	**115.6*****
H × C	0.7	0.3	0.2	0.0	0.2	2.3
H × N	3.0	0.3	0.3	0.1	0.8	0.6
C × N	0.6	**5.0***	1.2	1.6	**7.8****	**7.0***
H × C × N	0.8	0.7	0.7	1.9	0.3	<0.1
Basal part						
Block	**2.8***	**5.0****	**3.3***	**5.5*****	0.5	0.6
Herbivory (H)	0.2	0.2	<0.1	2.5	1.5	0.5
Connection (C)	0.6	0.4	0.2	2.6	2.7	1.0
Nitrogen (N)	**7.3***	0.6	0.1	**486.9*****	**323.8*****	**109.5*****
H × C	<0.1	2.7	2.4	0.1	0.1	1.2
H × N	1.2	0.1	<0.1	1.6	0.1	2.0
C × N	**4.5***	**5.5***	0.8	2.2	2.2	3.4
H × C × N	0.7	0.0	0.0	1.1	1.2	0.3

Numbers are ANOVA *F*-values. The degrees of freedom (*df*) for block effect is 5,35; the *df* values for foliar herbivory, clonal integration, nitrogen supply and their interactions are all 1,35. Values for which $P < 0.05$ are in bold.
Asterisks indicate significance level: no symbol, $P > 0.05$; *$P = 0.01–0.05$; **$P = 0.001–0.01$; and ***$P < 0.001$.

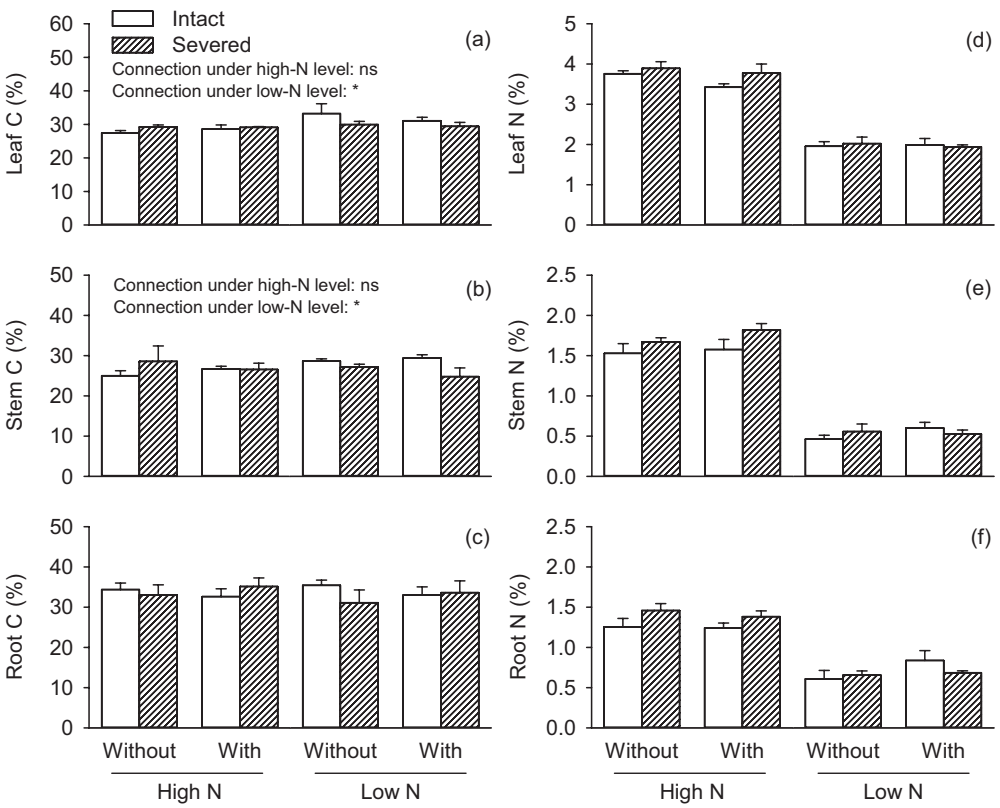

Figure 5: effects of foliar herbivory (with or without herbivory), clonal integration (intact or severed) and N supply (high or low) on the C and N concentrations of different plant organs from the basal parts of clonal fragments of *Alternanthera philoxeroides*. Error bars indicate means + SE. Symbols indicate significant differences between connection treatments at each N level, i.e. $P > 0.05^{ns}$; $P < 0.05$*.

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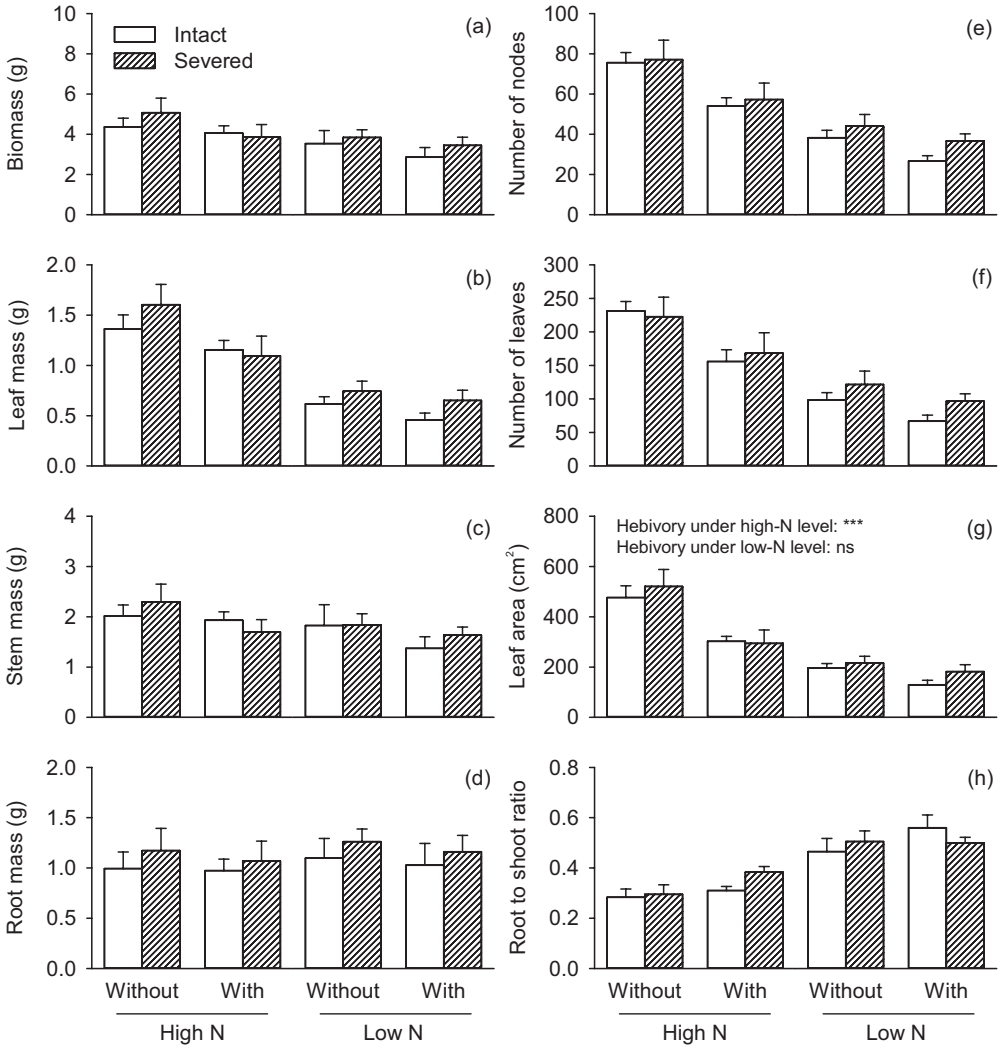

Figure 6: effects of foliar herbivory (with or without herbivory), clonal integration (intact or severed) and N supply (high or low) on the growth measures and biomass allocation of whole clonal fragments of *Alternanthera philoxeroides*. Error bars indicate means + SE. Symbols indicate significant differences between herbivory treatments at each N level, i.e. $P > 0.05^{ns}$; $P < 0.001$***.

regrowth ability of damaged ramets. Meanwhile, the developmental-programmed clonal integration to some extent mediated the physiological properties between individual ramets of *A. philoxeroides*, to facilitate the early growth and expansion of young ramets. However, since clonal integration may be a chronic response of *A. philoxeroides* against herbivory and the different intensity of herbivory may cause differences in strength of clonal integration, the timing and intensity of herbivory should be considered in future studies, to increase understanding of ecological significance of clonal integration under different herbivory conditions.

SUPPLEMENTARY MATERIAL

Supplementary material is available at *Journal of Plant Ecology* online.

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Conflict of interest statement. None declared.

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REFERENCES

Alpert P (1996) Nutrient sharing in natural clonal fragments of *Fragaria chiloensis*. *J Ecol* **84**:395–406.

Alpert P, Holzapfel C, Benson JM (2002) Hormonal modification of resource sharing in the clonal plant *Fragaria chiloensis*. *Funct Ecol* **16**:191–7.

Alpert P, Holzapfel C, Slominski C (2003) Differences in performance between genotypes of *Fragaria chiloensis* with different degrees of resource sharing. *J Ecol* **91**:27–35.

Alpert P, Mooney HA (1986) Resource sharing among ramets in the clonal herb, *Fragaria chiloensis*. *Oecologia* **70**:227–33.

Coley PD, Bryant JP, Chapin FS III (1985) Resource availability and plant antiherbivore defense. *Science* **230**:895–9.

D'Hertefeldt T, van der Putten WH (1998) Physiological integration of the clonal plant *Carex arenaria* and its response to soil-borne pathogens. *Oikos* **81**:229–37.

Dong BC, Alpert P, Zhang Q, *et al.* (2015) Clonal integration in homogeneous environments increases performance of *Alternanthera philoxeroides*. *Oecologia* **179**:393–403.

Dong BC, Fu T, Luo FL, *et al.* (2017) Herbivory-induced maternal effects on growth and defense traits in the clonal species *Alternanthera philoxeroides*. *Sci Total Environ* **605–606**:114–23.

Du J, Yu FH, Alpert P, *et al.* (2009) Arbuscular mycorrhizal fungi reduce effects of physiological integration in *Trifolium repens*. *Ann Bot* **104**:335–44.

Fornoni J, Núñezfarfán J, Valverde PL (2003) Evolutionary ecology of tolerance to herbivory: advances and perspectives. *Comments Theor Biol* **8**:643–63.

Fu JW, Shi MZ, Wang T, *et al.* (2016) Demography and population projection of flea beetle, *Agasicles hygrophila* (Coleoptera: Chrysomelidae), fed on alligator weed under elevated CO_2. *J Econ Entomol* **109**:1116–24.

Gao Y, Wang D, Xing F, *et al.* (2014) Combined effects of resource heterogeneity and simulated herbivory on plasticity of clonal integration in a rhizomatous perennial herb. *Plant Biol* **16**:774–82.

Gianoli E, Salgado-Luarte C (2017) Tolerance to herbivory and the resource availability hypothesis. *Biol Lett* **13**:20170120.

Gomez S, van Dijk W, Stuefer JF (2010) Timing of induced resistance in a clonal plant network. *Plant Biol* **12**:512–7.

Gomez S, Onoda Y, Ossipov V, *et al.* (2008) Systemic induced resistance: a risk-spreading strategy in clonal plant networks? *New Phytol* **179**:1142–53.

Hawkes CV, Sullivan JJ (2001) The impact of herbivory on plants in different resource conditions: a meta-analysis. *Ecology* **82**:2045–58.

Holm LG, Doll J, Holm E, *et al.* (1997) *World Weeds: Natural Histories and Distribution*. New York, NY: John Wiley and Sons.

Hosaka N, Kachi N, Kudoh H, *et al.* (2016) Compensatory growth of the clonal understory tree, *Asimina triloba,* in response to small-scale disturbances. *Plant Ecol* **217**:471–80.

Hutchings MJ, Wijesinghe DK (1997) Patchy habitats, division of labour and growth dividends in clonal plants. *Trends Ecol Evol* **12**:390–4.

Jia X, Pan XY, Li B, *et al.* (2009) Allometric growth, disturbance regime, and dilemmas of controlling invasive plants: a model analysis. *Biol Invasions* **11**:743–52.

Julien MH, Skarratt B, Maywald GF (1995) Potential geographical distribution of alligator weed and its biological control by *Agasicles hygrophila*. *J Aquat Plant Manage* **33**:55–60.

Lin HF, Alpert P, Zhang Q, *et al.* (2018) Facilitation of amphibious habit by physiological integration in the clonal, perennial, climbing herb *Ipomoea aquatica*. *Sci Total Environ* **618**:262–8.

Liu HD, Yu FH, He WM, *et al.* (2009) Clonal integration improves compensatory growth in heavily grazed ramet populations of two inland-dune grasses. *Flora* **204**:298–305.

Lu XM, Ding JQ (2012) History of exposure to herbivores increases the compensatory ability of an invasive plant. *Biol Invasions* **14**:649–58.

Lu XM, Shao X, Ding JQ (2014) No impact of a native beetle on exotic plant performance and competitive ability due to plant compensation. *Plant Ecol* **215**:275–84.

Lyu XQ, Zhang YL, You WH (2016) Growth and physiological responses of *Eichhornia crassipes* to clonal integration under experimental defoliation. *Aquat Ecol* **50**:153–62.

Outridge PM, Hutchinson TC (1990) Effects of cadmium on integration and resource allocation in the clonal fern *Salvinia molesta*. *Oecologia* **84**:215–23.

Pan XY, Geng YP, Zhang WJ, *et al.* (2006) The influence of abiotic stress and phenotypic plasticity on the distribution of invasive *Alternanthera philoxeroides* along a riparian zone. *Acta Oecol* **30**:333–41.

Pan XY, Jia X, Zeng J, *et al.* (2011) Stem tissue mass density is linked to growth and resistance to a stem-boring insect in *Alternanthera philoxeroides*. *Plant Spec Biol* **26**:58–65.

Pauliukonis N, Gough L (2004) Effects of the loss of clonal integration on four sedges that differ in ramet aggregation. *Plant Ecol* **173**:1–15.

R Core Team (2016) *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. https://www.R-project.org/ (31 October 2016, date last accessed).

Rasband WS (2017) *ImageJ 1.51j8*. Bethesda, MD: U.S. National Institutes of Health. http://imagej.nih.gov/ij/ (23 June 2017, date last accessed).

Rodríguez J, Calbi M, Roiloa SR, *et al.* (2018) Herbivory induced nonlocal responses of the clonal invader *Carpobrotus edulis* are not mediated by clonal integration. *Sci Total Environ* **633**:1041–50.

Roiloa SR, Rodríguez-Echeverría S, Freitas H, *et al.* (2013) Developmentally-programmed division of labour in the clonal invader *Carpobrotus edulis*. *Biol Invasions* **15**:1895–05.

Roiloa SR, Rodriguez-Echeverria S, Lopez-Otero A, *et al.* (2014) Adaptive plasticity to heterogeneous environments increases capacity for division of labor in the clonal invader *Carpobrotus edulis* (Aizoaceae). *Am J Bot* **101**:1301–8.

Rosario-Martinez HD (2015) *Post-Hoc Interaction Analysis. R Package Version 0.2–1* https://cran.r-project.org/web/packages/phia/ (17 May 2017, date last accessed).

Sainty G, McCorkelle G, Julien M (1998) Control and spread of alligator weed *Alternanthera philoxeroides* (Mart.) Griseb., in Australia: lessons for other regions. *Wetl Ecol Manag* **5**:195–201.

Schmid B, Puttick GM, Burgess KH, *et al.* (1988) Clonal integration and effects of simulated herbivory in old-field perennials. *Oecologia* **75**:465–71.

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Schooler S, Baron Z, Julien M (2006) Effect of simulated and actual symand reproduction. *Biol Control* **36**:74–9.

Song YB, Yu FH, Keser LH, *et al.* (2013) United we stand, divided we fall: a meta-analysis of experiments on clonal integration and its relationship to invasiveness. *Oecologia* **171**:317–27.

Spencer NR, Coulson JR (1976) The biological control of alligatorweed, *Alternanthera philoxeroides*, in the United States of America. *Aquat Bot* **2**:177–90.

Stuefer JF (1998) Two types of division of labour in clonal plants: benefits, costs and constraints. *Perspect Plant Ecol* **1**:47–60.

Stuefer JF, Gomez S, van Molken T (2004) Clonal integration beyond resource sharing: implications for defence signalling and disease transmission in clonal plant networks. *Evol Ecol* **18**:647–67.

Sun Y, Ding JQ, Frye MJ (2010) Effects of resource availability on tolerance of herbivory in the invasive *Alternanthera philoxeroides* and the native *Alternanthera sessilis*. *Weed Res* **50**:527–36.

Sun Y, Ding JQ, Ren MX (2009) Effects of simulated herbivory and resource availability on the invasive plant, *Alternanthera philoxeroides* in different habitats. *Biol Control* **48**:287–93.

Tewari S, Buonaccorsi JP, Averill AL (2014) Physiological integration plays key role in cranberry (Ericales: Ericaceae) for tolerance of damage by *Dasineura oxycoccana* (Diptera: Cecidomyiidae). *Environ Entomol* **43**:75–82.

Wang P, Alpert P, Yu FH (2017a) Clonal integration affects allocation in the perennial herb *Alternanthera philoxeroides* in N-limited homogeneous environments. *Folia Geobot* **502**:303–15.

Wang P, Li H, Pang XY, *et al.* (2017b) Clonal integration increases tolerance of a phalanx clonal plant to defoliation. *Sci Total Environ* **593–594**:236–41.

Wang YJ, Müller-Schärer H, van Kleunen M, *et al.* (2017c) Invasive alien plants benefit more from clonal integration in heterogeneous environments than natives. *New Phytol* **216**:1072–8.

Wang N, Yu FH, Li PX, *et al.* (2008) Clonal integration affects growth, photosynthetic efficiency and biomass allocation, but not the competitive ability, of the alien invasive *Alternanthera philoxeroides* under severe stress. *Ann Bot* **101**:671–8.

Wang N, Yu FH, Li PX, *et al.* (2009) Clonal integration supports the expansion from terrestrial to aquatic environments of the amphibious stoloniferous herb *Alternanthera philoxeroides*. *Plant Biol* **11**:483–9.

Wilson JRU, Yeates A, Schooler S, *et al.* (2007) Rapid response to shoot removal by the invasive wetland plant, alligator weed (*Alternanthera philoxeroides*). *Environ Exp Bot* **60**:20–5.

Wise MJ, Abrahamson WG (2007) Effects of resource availability on tolerance of herbivory: a review and assessment of three opposing models. *Am Nat* **169**:443–54.

Xu CY, Schooler SS, van Klinken RD (2010) Effects of clonal integration and light availability on the growth and physiology of two invasive herbs. *J Ecol* **98**:833–44.

Xu CY, Zhang WJ, Fu CZ, *et al.* (2003) Genetic diversity of alligator weed in China by RAPD analysis. *Biodivers Conserv* **12**:637–45.

Xu L, Yu FH, van Drunen E, *et al.* (2012) Trampling, defoliation and physiological integration affect growth, morphological and mechanical properties of a root-suckering clonal tree. *Ann Bot* **109**:1001–8.

Ye WH, Li J, Cao HL, *et al.* (2003) Genetic uniformity of *Alternanthera philoxeroides* in South China. *Weed Res* **43**:297–302.

You WH, Yu D, Xie D, *et al.* (2014) The invasive plant *Alternanthera philoxeroides* benefits from clonal integration in response to defoliation. *Flora* **209**:666–73.

Yu FH, Dong M, Krusi B (2004) Clonal integration helps *Psammochloa villosa* survive sand burial in an inland dune. *New Phytol* **162**:697–704.

Yu FH, Wang N, Alpert P, *et al.* (2009) Physiological integration in an introduced, invasive plant increases its spread into experimental communities and modifies their structure. *Am J Bot* **96**:1983–9.

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